



Indian Institute of Technology Guwahati

BSc. (Hons.) Data Science & Artificial Intelligence
Online Degree Programme

INTERNSHIP/TERM PROJECT

Does Machine Learning Beat the Credit Scorecard?

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THE PROBLEM

Probability of Default: the Highest-Stakes Model in Banking

The task: estimate the probability that a loan applicant defaults, using only information available on the day of application.

Why it matters: PD models drive accept/reject decisions, pricing, and, under Basel and IFRS 9, regulatory capital and provisions.

The status quo: banks still use logistic-regression scorecards, a forty-year-old technology, because they are stable and explainable.

The ML claim: benchmarking papers report boosted trees dominate, but they score models almost exclusively by AUC.

The question: does the ML advantage survive the criteria a lender actually needs?

1. Discrimination

Can the model rank bad borrowers above good ones? (AUC, KS)

2. Calibration

Can the predicted PD be taken at face value? (Brier, ECE, reliability)

3. Profit

Does the model make money at its accept/reject cutoff?

A Reproducible Python Stack, CPU Only

Language

Python 3.13

Data

pandas
NumPy
kagglehub

Models

scikit-learn
XGBoost
LightGBM
SciPy

Workflow

Jupyter
VS Code
Git / GitHub
LaTeX

Everything is scripted: one command reproduces the full study, about 18 minutes on a laptop; all seeds fixed, one shared vintage split.

DATASET

Lending Club Personal Loans, Engineered Against Leakage

621,022

36-month loans, 2007-2015

151 → 23

columns after leakage whitelist

12.6 → 14.9%

default rate, train → test vintage

30,000

UCI Taiwan clients (robustness)

Source: public Lending Club export (Kaggle); 36-month loans only, so every loan matures before the data ends and no outcome is censored.

Target: charged-off vs fully paid; loans still in flight are excluded.

The leakage filter is the hard work: the raw export has 151 columns, most of them post-origination servicing fields (payments received, recoveries, hardship flags) that trivially predict the outcome. I kept an explicit whitelist of 23 origination-time fields.

Out-of-time split: train on 2007-2013 vintages, calibrate on 2014, test on 2015, exactly how a deployed model meets the future. The rising default rate across vintages is the drift a real model faces.

One Pipeline, Four Models, Identical Information

The creditrisk package

data loaders + leakage whitelist + vintage split

woe / scorecard WoE binning, IV screen, logistic, points scaling

models XGBoost, LightGBM, MLP on a shared design matrix

calibration Platt + isotonic, fit on the 2014 vintage

metrics AUC, KS, Brier, ECE, DeLong test, bootstrap CIs

decision accept/reject profit sweep over PD cutoffs

pipeline end-to-end orchestration, 5 seeds per ML model

The baseline is honest: an industry-style weight-of-evidence scorecard, not a strawman logistic regression on raw inputs.

Same information for everyone: all four models see the identical 23 origination-time features; preprocessing differences cannot explain gaps.

Calibration is separated from fitting: every model is recalibrated the same way on the validation vintage, mimicking bank practice.

Profit uses real cash flows: repaid loans earn their actual interest; charged-off loans lose 65% of principal.

Three Questions, Each With Its Own Metrics

Discrimination: AUC, KS

AUC: probability a random defaulter is ranked riskier than a random good borrower. KS: maximum distance between the two score distributions. Both ignore whether the PD values themselves are right.

Calibration: Brier, ECE, reliability

Among loans assigned PD near q , do a fraction q actually default? Brier: mean squared error of the probability. ECE: average gap between predicted PD and observed default rate over 10 bins.

Decision quality: profit, with significance

Realized profit per 1,000 applicants across accept/reject cutoffs, from actual cash flows. Significance throughout: DeLong test for correlated AUCs, 1,000-resample bootstrap CIs.

Why three? A model can gain AUC while its probabilities become less trustworthy; 'best model' is undefined until the question is fixed.

Discrimination: the Trees Win, the Neural Net Loses

Model	AUC	KS	ΔAUC vs scorecard (DeLong)
Scorecard	0.6767	0.2570	baseline
XGBoost	0.6850	0.2701	+0.85 pts, $p < 10^{-54}$
LightGBM	0.6843	0.2693	+0.72 pts, $p < 10^{-48}$
MLP	0.6716	0.2542	-0.17 pts, $p = 0.024$ (worse)

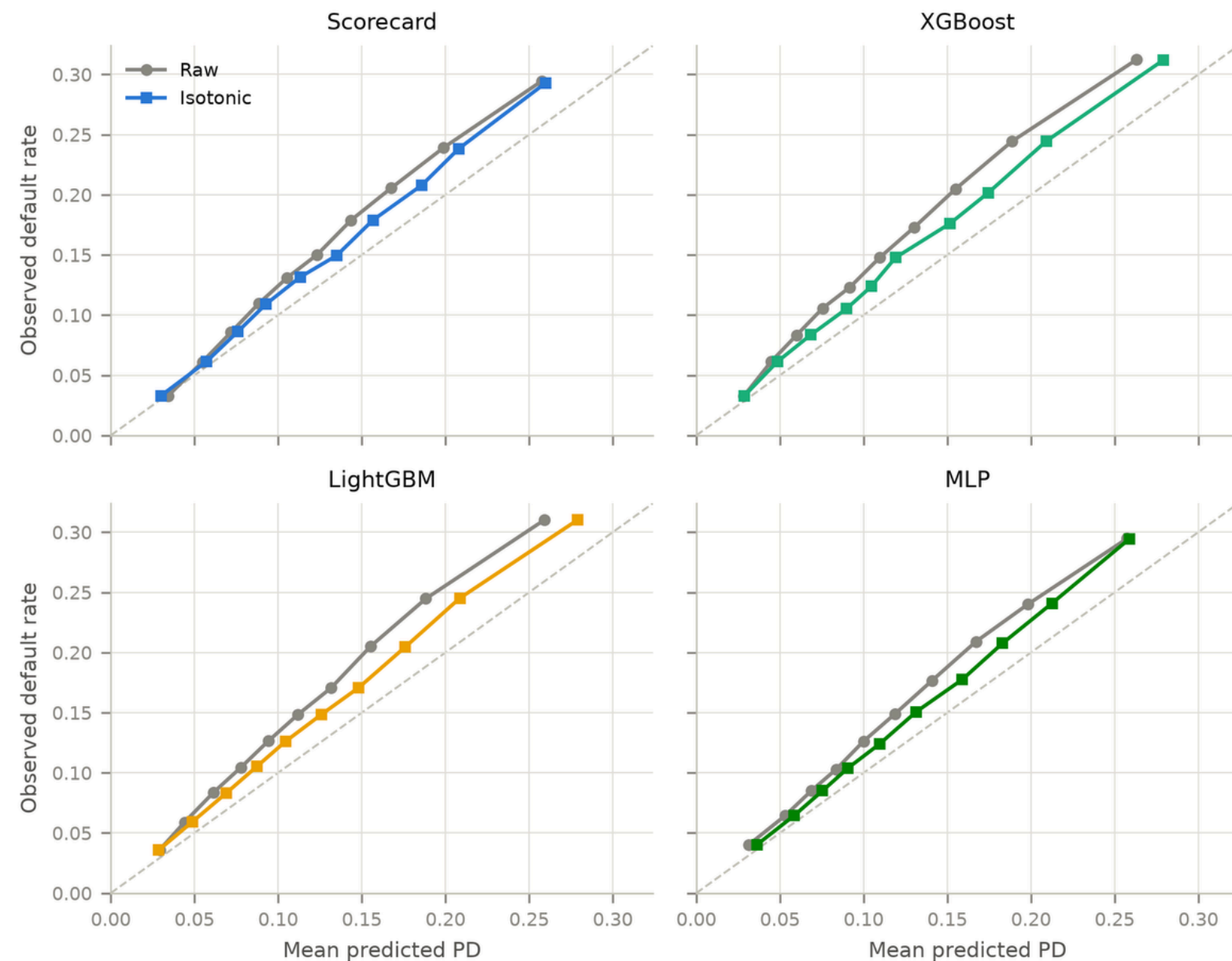
Statistically decisive: 283,026 test loans give enormous power; the boosted-tree edge is unambiguous.

Economically modest: +0.85 AUC points, not the +8 to +15 sometimes quoted from random-split studies.

Deep learning does not pay: the MLP is significantly worse than the forty-year-old scorecard, on tabular credit data.

Why so close? The features include the platform's own risk pricing (grade, interest rate), and the out-of-time split denies flexible models the flattery of a random split.

Calibration: Every Raw Model Under-Predicts the Future



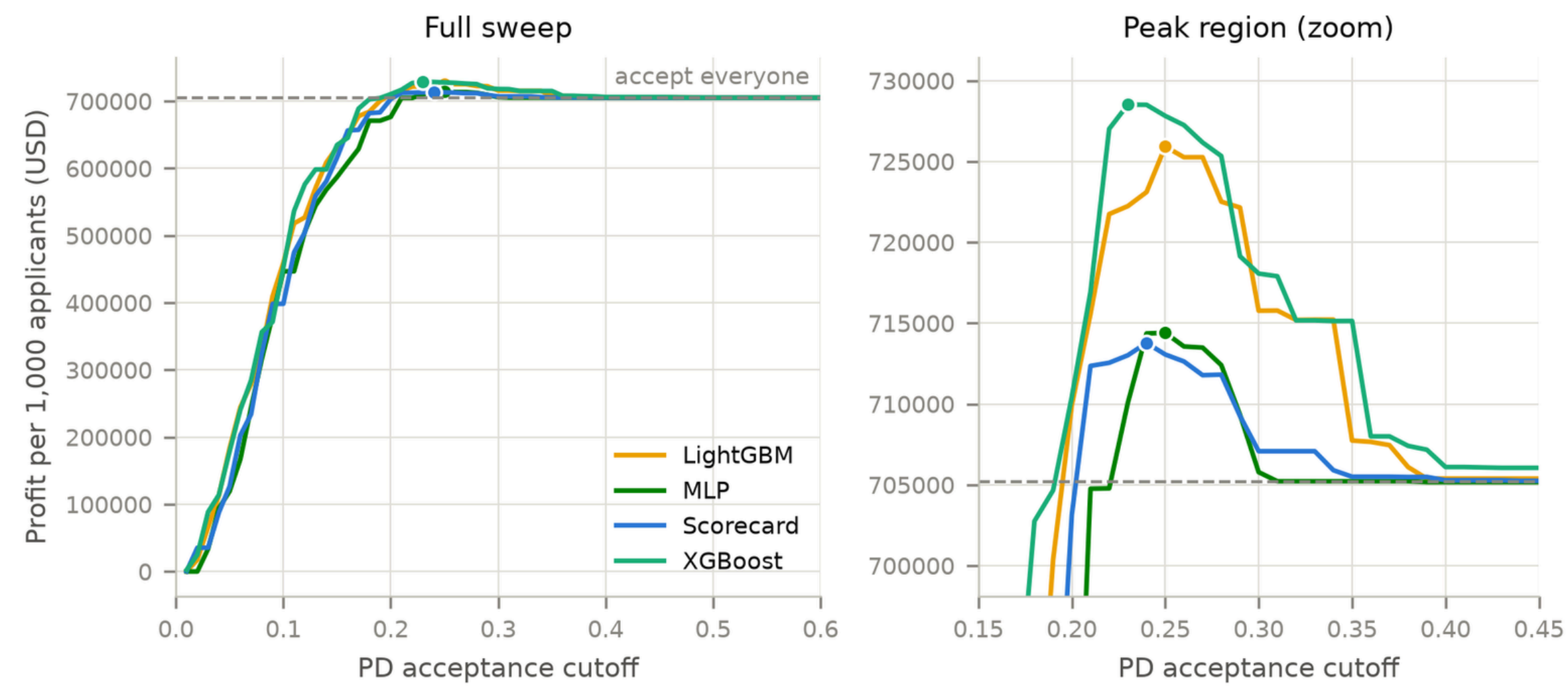
Read the diagonal: a calibrated model's curve lies on it. Every raw curve (grey) sits above: the 2015 vintage defaulted more than the training years, so all models under-predict PD.

Raw trees are worst: ECE 0.034 vs the scorecard's 0.025; margin-maximizing ensembles are not probability models.

Isotonic recalibration helps everyone but cannot anticipate drift that was not yet visible in 2014.

The scorecard stays best-calibrated: ECE 0.0175 vs XGBoost's 0.0214 after identical recalibration. The trees keep the better Brier score, which also rewards discrimination.

Profit: the AUC Gap Becomes Money at the Cutoff



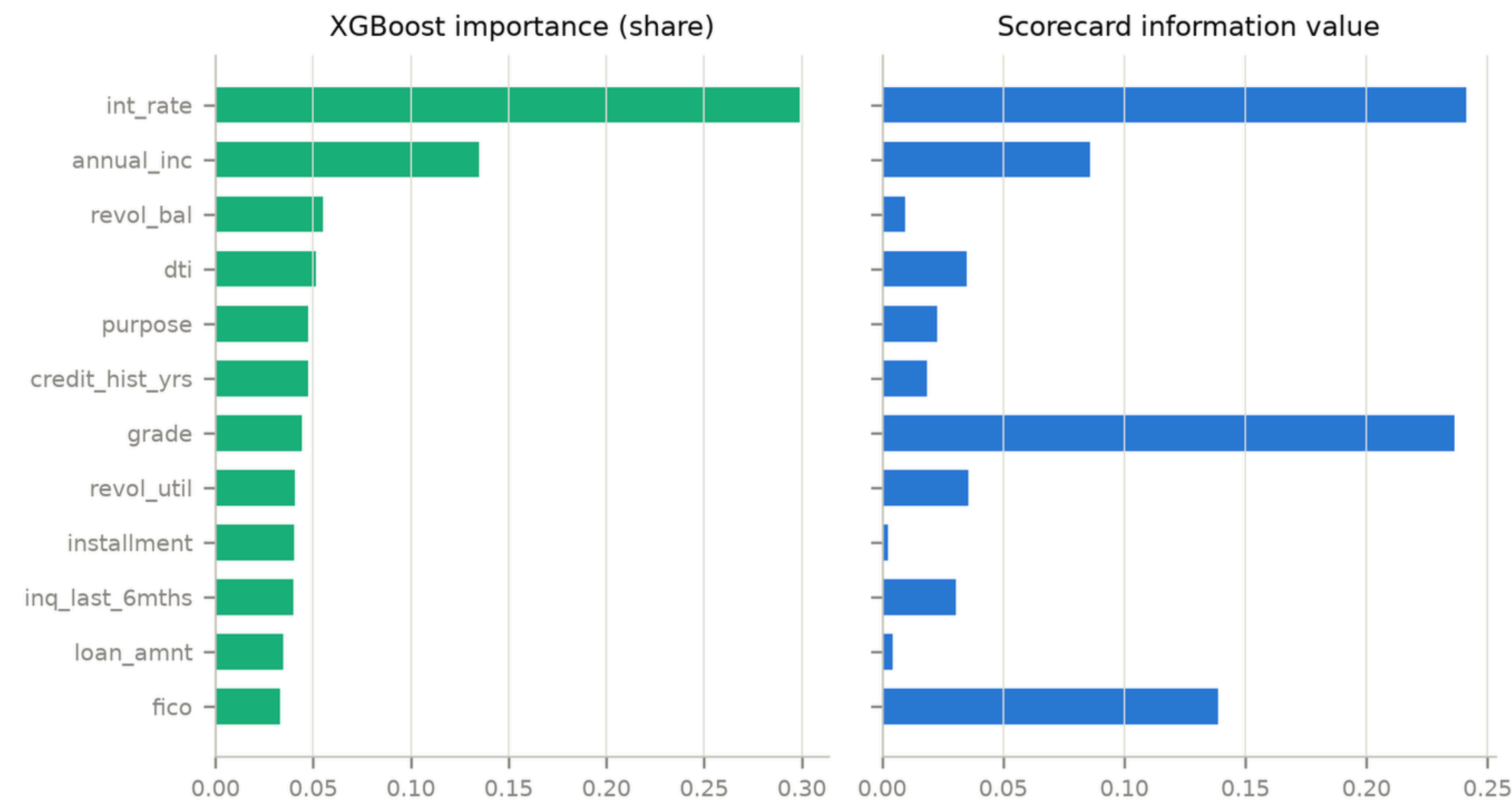
Accepting everyone already earns \$705 per applicant; interest income dominates this portfolio.

Scorecard: reject the riskiest 6.7% for +1.2% profit.

XGBoost: reject 9.3% for +3.3%, that is +\$14,700 per 1,000 applicants over the scorecard (+2.1%).

Scale decides adoption: a modest edge worth millions at bank volume, and nothing at credit-union volume.

What Did the Models Learn? Mostly the Same Thing



Both families agree that the platform's own pricing (interest rate, grade) carries the most signal; rank correlation 0.56.

They disagree in encoding: the scorecard leans on sub-grade and FICO bins; XGBoost prefers the continuous interest rate and income.

Interpretability trade-off: the scorecard is a readable sum of points per applicant; the ensemble needs post-hoc explanation tools.

Same Ranking on an Independent Dataset

UCI Taiwan credit-card default (30k clients)

XGBoost 0.788 vs scorecard 0.772 AUC: +1.50 pts, $p = 2.8 \times 10^{-6}$

MLP 0.766 again below the scorecard

Scorecard again best ECE after identical recalibration

Honest limitations

Accepted loans only: all models inherit Lending Club's own selection; reject inference is an open problem.

Simple profit model: undiscounted, fixed LGD of 0.65, no partial recoveries or prepayment.

Platform grades kept as features: realistic for an investor, but understates what ML could add on raw bureau data.

One market, one test vintage: 2015 was benign; a recession vintage could reorder the calibration findings.

My scorecard is unconstrained: expert binning with monotonicity constraints would likely close part of the gap.

A Qualified Yes: ML Wins, but Narrowly, and Only the Trees

Boosted trees beat the scorecard on all three criteria: discrimination (+0.85 AUC pts, $p < 10^{-54}$), Brier after identical recalibration, and profit (+2.1% at the optimal cutoff).

But the margins are modest and uneven: the MLP loses to the scorecard outright, and the scorecard stays the best-calibrated model throughout.

What I learned: leakage control was the highest-leverage work in the project; the model ranking changes with the question asked; and vintage drift is why PD models are monitored, not trusted.

Future work: reject inference, default-timing survival models, discounted cash flows, monotonic gradient boosting, a recession-vintage stress test.